

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION**

81 Higuera Street, Suite 200
San Luis Obispo, California 93401-5427

ORDER NO. 94-82

**WATER RECLAMATION REQUIREMENTS
FOR
MONTEREY REGIONAL WATER POLLUTION CONTROL AGENCY
(PRODUCER)
WATER RECLAMATION PROJECT,
MONTEREY COUNTY**

The California Regional Water Quality Control Board, Central Coast Region (hereafter Board), finds:

1. John Klein, Senior Engineer, filed a Report of Waste Discharge on April 12, 1994, in accordance with Section 13522.5 of the California Water Code. The report was filed on behalf of Monterey Regional Water Pollution Control Agency (MRWPCA) for authorization to produce reclaimed water for delivery to the Castroville Irrigation Project and also for landscape irrigation and wash down water at the Regional Treatment Plant. The information supports a request for issuance of producer/user water reclamation waste discharge requirements.
2. Monterey Regional Water Pollution Control Agency (hereafter Discharger), 5 Harris Court, Monterey, CA, proposes to own and operate water reclamation facilities located at the Monterey Regional Wastewater Treatment Plant, owned and operated by the Discharger. Location of the proposed facilities are as shown on Attachment "A" of this Order.
3. Up to 29.6 million gallons per day (average dry weather flow) of secondary treated effluent from the Monterey Regional Wastewater Treatment Plant is proposed to be treated by the reclamation facility. Treatment will consist of chemical coagulation, flocculation, filtration and disinfection. Filter backwash will be returned to the secondary plant for disposal. Solid wastes generated from the reclamation facility will be disposed at the Marina Landfill. An onsite pond will store up to 80 acre feet (26 million gallons) of reclaimed water.
4. The Primary User of reclaimed water will be the Monterey County Water Resources Agency. The MCWRA will control the proposed water distribution system that will convey reclaimed water to the approximately 12,100 acres of farmland in the Castroville area as shown on Attachment "B". Separate waste discharge requirements for use of reclaimed water will be issued to Monterey County Water Resources Agency.
5. Reclaimed water failing to meet the required standards for food crop spray irrigation will be diverted back to the Regional Treatment Plant headworks, reclamation plant influent, or to the ocean outfall. Secondary effluent in excess of that diverted for reclamation will be discharged through the existing ocean outfall. Outfall discharges will continue to be regulated by NPDES Permit No. CA0048551, Order No. 92-64.
6. The Discharger is currently permitted to discharge up to 29.6 million gallons-per-day (average dry weather flow) of secondary treated effluent to the Pacific Ocean through an ocean outfall/diffuser system.
7. These waste discharge requirements are being issued to regulate reclaimed water production and reuse at the regional treatment plant. This type of discharge has never been regulated by the Board for this facility.
8. The Water Quality Control Plan, Central Coastal Basin (Basin Plan), was adopted by the Board on November 17, 1989 and approved by

the State Board on August 16, 1990. The Basin Plan incorporates statewide plans and policies by reference and contains a strategy for protecting beneficial uses of State waters.

9. Present and anticipated beneficial uses of groundwater in the vicinity of the discharge include:
 - a. Domestic and Municipal Supply;
 - b. Agricultural Supply; and,
 - c. Industrial Supply.
10. State Department of Health Services' criteria for use of reclaimed water is in the California Code of Regulations, Title 22, Division 4, Chapter 3. The Board has consulted with the State Department of Health Services regarding the regulation of this discharge.
11. The Monterey County Water Resources Agency and the Monterey Regional Water Pollution Control Agency, certified a final Environmental Impact Report on April 7, 1992, and April 27, 1994, respectively, in accordance with the California Environmental Quality Act (Public Resources Code, Section 21000, et.seq.) and the California Code of Regulations.
12. Discharge of waste is a privilege, not a right, and authorization to discharge is conditional upon the discharge complying with provisions of Division 7 of the California Water Code and any more stringent effluent limitations necessary to implement water quality control plans, to protect beneficial uses, and to prevent nuisance. Compliance with this Order should assure this and mitigate any potential adverse changes in water quality due to the discharge.
13. On July 19, 1994, the Board notified the Discharger and interested agencies and persons of its intent to issue water reclamation requirements for the discharge and has provided them with a copy of the proposed Order and an opportunity to submit written views and comments.
14. After considering all comments pertaining to this discharge, during a public hearing on

September 9, 1994, this Order was found consistent with the above findings.

IT IS HEREBY ORDERED, pursuant to authority in Section 13523 of the California Water Code, the Monterey Regional Water Pollution Control Agency, its agents, successors, and assigns, may produce and use reclaimed water at the above described wastewater treatment plant providing compliance is maintained with the following:

(Note: other prohibitions and conditions, definitions, and the method of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated January 1984. Applicable paragraphs are referenced in paragraph C.2. of this Order.)

Requirements specified in the proposed Orders are based on staff's professional judgement and the following documents:

- A = California Code of Regulations, Title 22
- B = Basin Plan
- C = Administrative Procedures Manual (SWRCB)
- D = Draft Guidelines for Reclaimed Water Use (DHS)
- E = Porter Cologne Water Quality Control Act

Throughout the proposed Order and Monitoring and Reporting Program footnotes are included to indicate the source of specified requirements. Requirements not referenced are based on professional judgement.

A. PROHIBITIONS

1. Discharge of reclaimed water to areas other than landscape irrigation areas and for use as wash down water at the regional treatment plant and Finding No. 4 is prohibited.*^D
2. Discharge of any wastes, including overflow, bypass, and overspray from treatment, storage, or transport to adjacent drainageways or adjacent properties is prohibited.

3. Discharge of reclaimed water for landscape irrigation not in compliance with Title 22, Division 4, Chapter 3, Article 4, Section 60313 of the California Code of Regulations (adequately oxidized, coagulated, clarified, filtered, disinfected wastewater, or a wastewater treated by a sequence of unit processes that will assure an equivalent degree of treatment and reliability), is prohibited.^A

*For the purposes of this prohibition, insignificant amounts of runoff from the extreme peripheries of spray irrigated areas shall not be considered a violation of this Order provided that the runoff is minimal and that the User utilizes good irrigation practices.

B. RECLAMATION SPECIFICATIONS

1. Daily average dry weather flow over each month shall not exceed 29.6 million gallons.
2. Reclaimed water shall not contain constituents in excess of the following:

<u>Constituent</u>	<u>Unit of Measurement</u>	<u>Mean</u>	<u>Maximum</u>
Biological Oxygen Demand, 5-Day	mg/l	10**	25
Turbidity***	NTU	2	5
Total Non-Filterable Residue (Suspended Solids)	mg/l	10**	25
Settleable Solids	ml/l/hr	-	0.1
Total Dissolved Solids	mg/l	-	1500

**Compliance shall be determined from the five most recent samples

***Limits to be met following filtration. Maximum limit shall not be exceeded at any time.

3. For spray irrigation to crops, the median number of total coliform organisms in reclaimed water shall not exceed 2.2 per 100 milliliters, as determined from the bacteriological results of the last 7 days for which analyses have been completed, and the number of total coliform organisms shall not exceed 23 per 100 milliliters in any sample.
4. Reclaimed water shall not have a pH less than 6.5 or greater than 8.4.
5. Total chlorine residual in reclaimed water shall equal or exceed 5 mg/l, as measured within the chlorine contact zone at the end of the chlorine contact chamber.
6. Use of reclaimed water shall cease and all wastewater shall be diverted immediately to the Regional Treatment Plant ocean outfall, existing treatment plant headworks, or reclamation plant influent, if Reclamation Specifications are violated or threaten to be violated.

7. Irrigation with reclaimed water shall occur at a time and in a manner to prevent or minimize contact with reclaimed water and to allow irrigated areas maximum opportunity to dry before use. Drinking fountains shall be protected from direct or windblown spray of reclaimed water.^D
8. All reclamation areas with public access shall be posted to warn the public that reclaimed water is being stored or used.^D
9. Personnel involved in producing, transporting or using reclaimed water shall be informed of possible hazards associated with contact or use of reclaimed water.^D
10. Tank trucks used for transporting reclaimed water shall be appropriately labeled and shall not leak.^D
11. Reclaimed water valves, outlets etc. shall be marked to differentiate reclaimed water facilities from potable water facilities. Proper backflow and cross-connection protection for domestic water services and irrigation wells shall be provided.^D
12. Reclaimed water valves, outlets, quick couplers and sprinklers shall be of a type, or secured in a manner, that permits operation only by authorized personnel. Use or installation of hose bibs on the reclaimed water system shall not be permitted.^D
13. Reclaimed water shall be applied at a rate and volume not to exceed vegetative demand and soil moisture holding conditions. Special precautions must be taken to prevent clogging of spray nozzles, over watering and ponding, and to minimize runoff. Pipelines shall be maintained to prevent leaks.^D

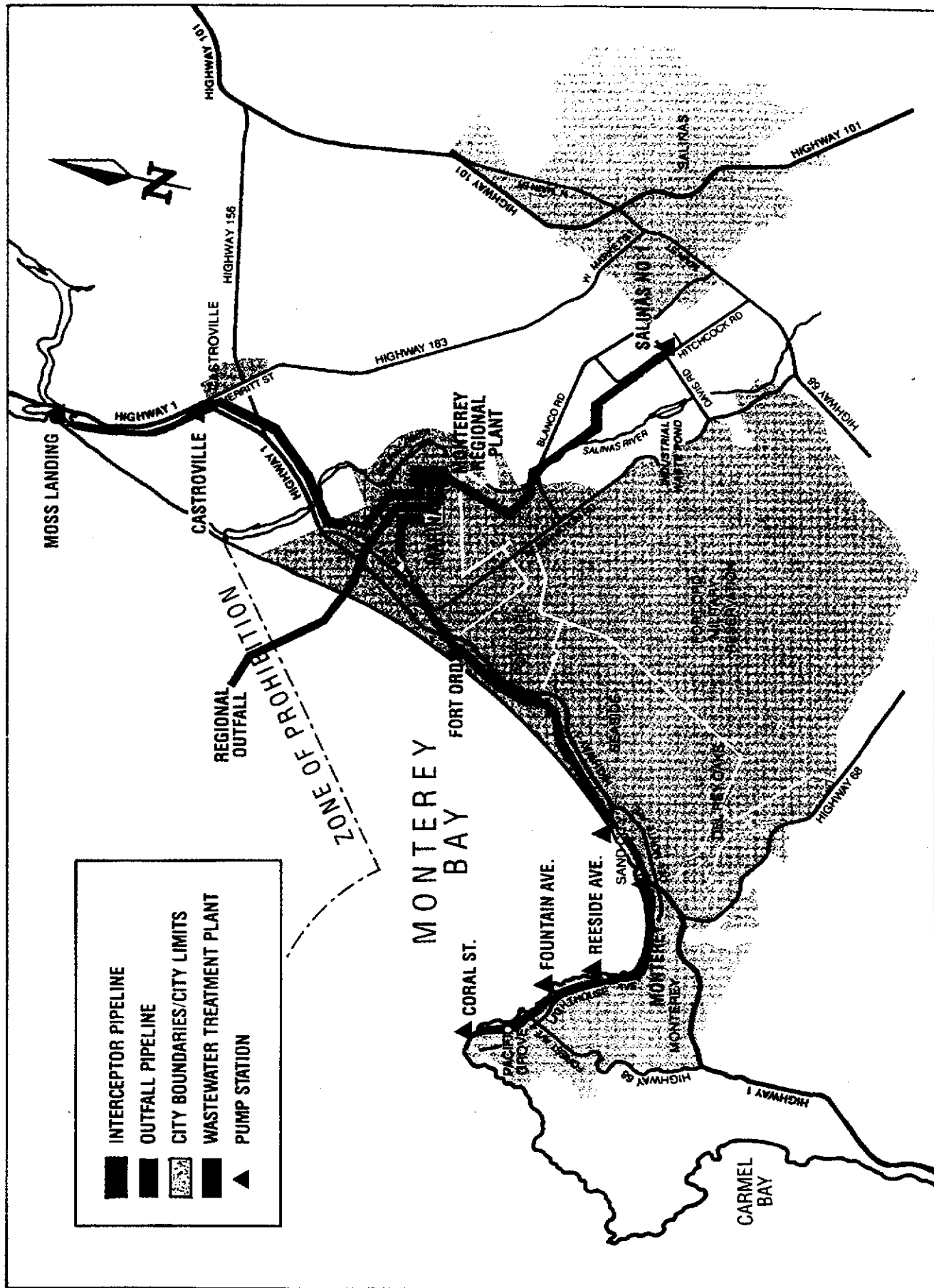
14. Reclaimed water shall not be used for irrigation during periods of extended rainfall and/or runoff.
15. Reclaimed water systems shall be inspected daily, according to Monitoring and Reporting Program No. 94-82, to assure proper operation, absence of leaks and absence of illegal connections.
16. Additional sites proposed for reclaimed water use require written approval by the Executive Officer.

C. PROVISIONS

1. Discharger shall comply with "Monitoring and Reporting Program No. 94-82," as specified by the Executive Officer.
2. Discharger shall comply with all items of the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated January, 1984; except A.11., A.14., A.15.; C.4., and D.1.
3. Pursuant to Title 23, Division 3, Chapter 9, of the California Code of Regulations, the Discharger must submit a report to the Executive Officer not later than March 10, 1999, addressing:
 - a. Whether there will be changes in the continuity, character, location, or volume of the discharge; and,
 - b. Whether, in their opinion, there is any portion of the Order that is incorrect, obsolete, or otherwise in need of revision.

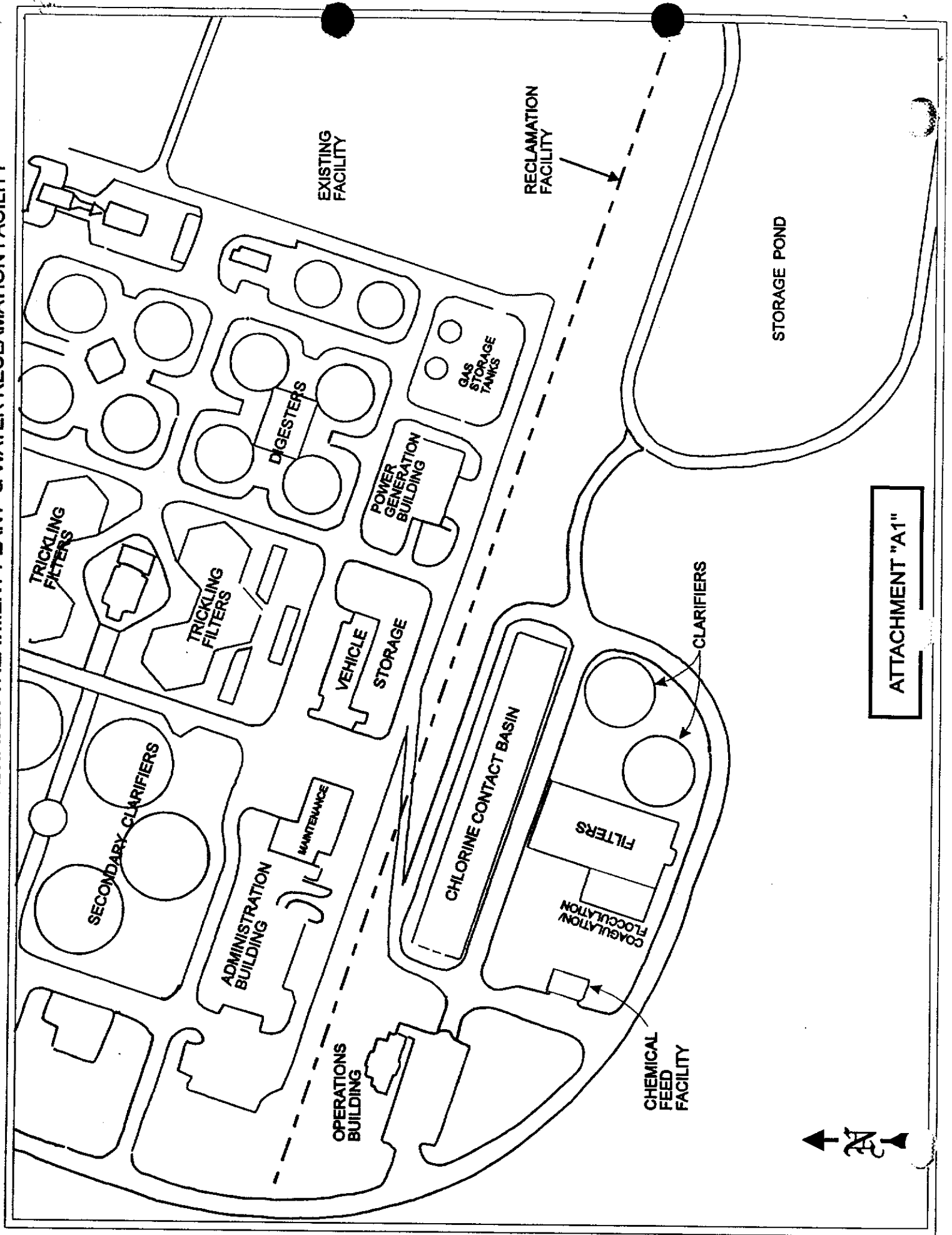
I, **ROGER W. BRIGGS**, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Coast Region, on September 9, 1994.


EXECUTIVE OFFICER

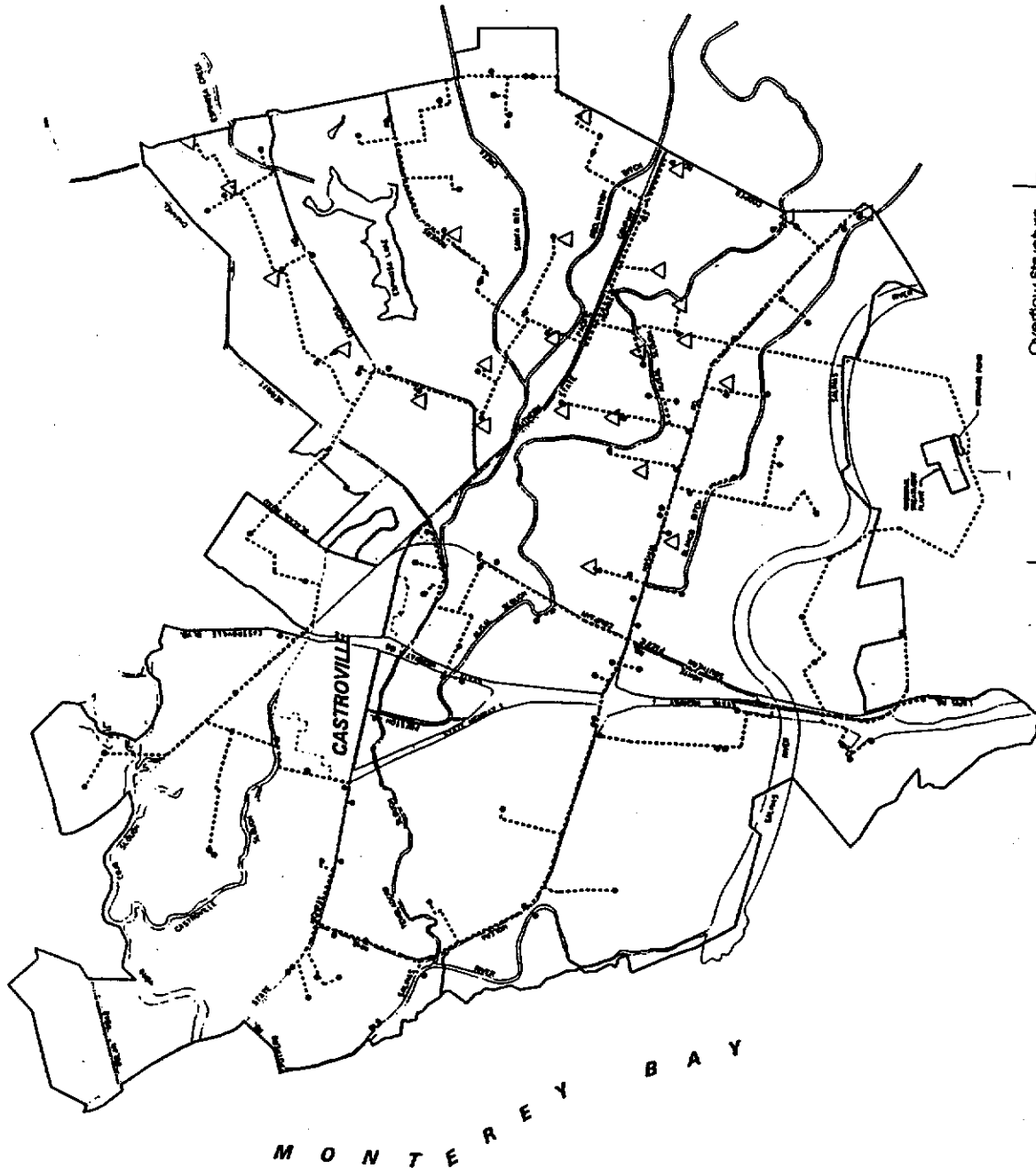


ATTACHMENT "A"

MONTEREY REGIONAL WASTEWATER TREATMENT PLANT & WATER RECLAMATION FACILITY



ATTACHMENT "A1"



MONTEREY COUNTY WATER RESOURCES AGENCY
 CASTROVILLE SEAWATER INTRUSION PROJECT
 PROPOSED PIPELINE ALIGNMENT
 FIGURE 1

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION**

**MONITORING AND REPORTING PROGRAM NO. 94-82
FOR
MONTEREY REGIONAL WATER POLLUTION CONTROL AGENCY
WATER RECLAMATION PROJECT
MONTEREY COUNTY**

Staff chose all constituents, sample types, and sampling and analyzing frequencies in Monitoring and Reporting Program (MRP) No. 94-82 by employing professional judgement to specify the monitoring parameters needed to ensure reclaimed water is adequately treated prior to delivery. The MRP's specifications are consistent with other programs for similar reclaimed wastewater treatment systems.

Reclaimed Water Monitoring

A sampling station shall be established where representative samples of the reclaimed water can be obtained at the end of disinfection and during reclaimed water production. The following shall constitute the reclaimed water monitoring program:

<u>Parameter</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Minimum Sampling and Analyzing Frequency</u>
Daily Flow*	gal/day	Metered	Daily
Maximum Daily Flow*	gal/day	-----	Monthly
Mean Daily Flow*	gal/day	Calculated	Monthly
Turbidity**	NTU	Metered	Continuous
Chlorine Residual***	mg/l	Metered	Continuous
Total Coliform Organisms	MPN/100 ml	Grab	Daily
Settleable Solids	ml/l/hr	Grab	Daily
pH	pH units	Grab	Daily
Biochemical Oxygen Demand, 5-day	mg/l	24-hr Composite	Weekly
Total Non-Filterable Residue (Suspended Solids)	mg/l	24-hr Composite	Weekly
Total Dissolved Solids	mg/l	24-hr Composite	Monthly

- * Flow shall be metered at the distribution system pump station to provide a record of the quantity of reclaimed water used each day (per normal irrigation period).
- ** Report daily maximum and daily mean values. Turbidity samples may be obtained anywhere in the treatment process following filtration.
- *** Report daily maximum and daily minimum values prior to discharge and at the end of the chlorine contact chamber. Compliance shall be determined by daily minimum values measured within the chlorine contact zone at the end of the chlorine contact chamber.

Reporting

Reports shall be submitted by the 20th day of the month and shall include (for previous month):

1. Results of reclaimed water monitoring;
2. A summary of operational problems, plant and equipment malfunctions, and any diversion of reclaimed water to the ocean outfall or back to the Regional Plant.
3. A record of equipment or process failures initiating an alarm, as well as, any corrective and preventative measures taken;
4. A summary of observations made during inspections, including description of any modification performed on the irrigation system and any employee training; and,
5. A statement that irrigation operations and reclaimed water use are in conformance with Order No. 94-82.

ORDERER BY


EXECUTIVE OFFICER

September 9, 1994

Date